

Introduction

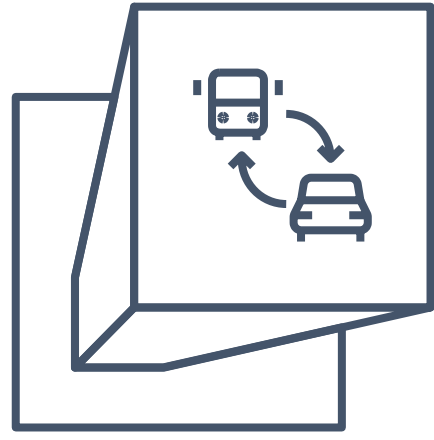
Mobility as a Service (MaaS)

MaaS is emerging as a solution that integrates multiple transportation mode



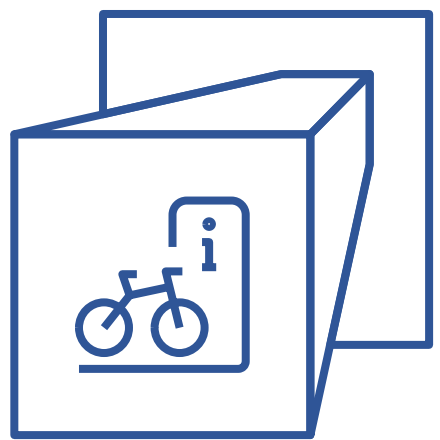
Seamless multimodal connectivity

Seamless multimodal connectivity enhances urban mobility significantly



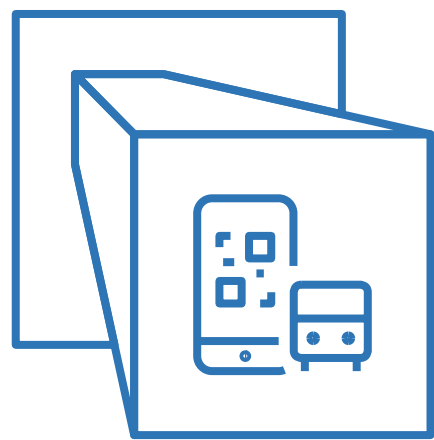
Bike-sharing(BS) System

Bike serves as a feeder, bridging first- and last-mile gaps



Integrated transport system

Integration of Bike-sharing(BS) and Public Transportation(PT) could led to significant improvements in urban mobility



Literature Review

Study	Key Focus
Lee et al., 2019a	Evaluate how physical station attributes influence transfer efficiency within PT system
Nielsen et al., 2021	Investigate how walking conditions and vertical accessibility affect the overall user experience during multimodal transfers
Becker et al., 2020	To assess the environmental impact of integrating BS with PT, particularly in reducing car dependency and emissions
Kosmidis & Müller-Eie, 2024	To explore how integrating BS into PT networks can enhance overall urban mobility and system-wide efficiency
Matowicki et al., 2022	Examine the role of BS systems in improving access to PT, especially for first- and last-mile connections

Although previous studies have provided valuable insights into transfer efficiency, research on evaluating the efficiency of transfers between PT and BS systems remains limited

This study aims to assess the efficiency of the overall transfer process from PT to BS services

Research Methodology

Proposed Network DEA Model

Unlike conventional Network DEA models, it optimizes PT trips in Stage 1 and uses them as fixed inputs in Stage 2, avoiding feedback and ensuring structural consistency.

Stage 1 : Public Transportation Efficiency Evaluation

Inputs	# of Bus Routes and Stops, Subway Lines, Transfer Distance(m)
Outputs	PT trips
Equation (1) ~ (4)	Evaluate how efficiently PT infrastructure generates passenger demand
$\sum_{j=1}^J x_m^j \lambda^j \leq x_m^k, \forall m$	(1)
$\sum_{j=1}^J z_p^j \lambda^j \geq \phi^k z_p^k, \forall p$	(2)
$\sum_{j=1}^J \lambda^j = 1$	(3)
$z_p^k = \phi^k z_p^k$	(4)

Stage 2 : Bike-Sharing Efficiency Evaluation

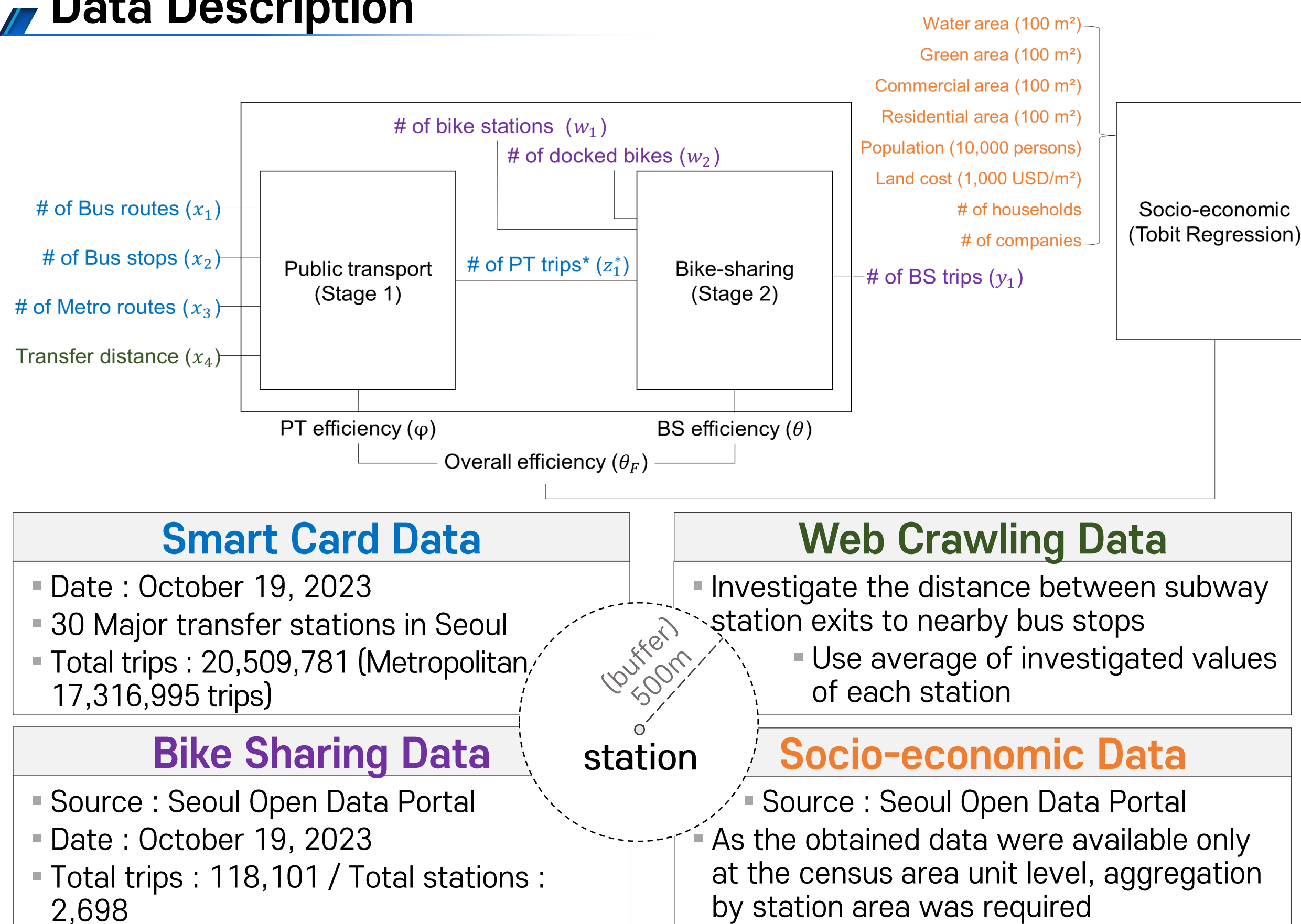
Inputs	Optimized PT trips($z_p^k = \phi^k z_p^k$), Bike Stations, # of Docked Bikes
Outputs	BS trips
Equation (5) ~ (8)	Measure how well bike-sharing infrastructure converts PT demand into actual usage
$\sum_{j=1}^J z_p^j \mu^j \leq z_p^k, \forall p$	(5)
$\sum_{j=1}^J w_q^j \mu^j \leq w_q^k, \forall q$	(6)
$\sum_{j=1}^J y_n^j \mu^j \geq \theta^k y_n^k, \forall n$	(7)
$\sum_{j=1}^J \mu^j = 1$	(8)

Overall Efficiency

Harmonic mean of Stage 1 and Stage 2 efficiency

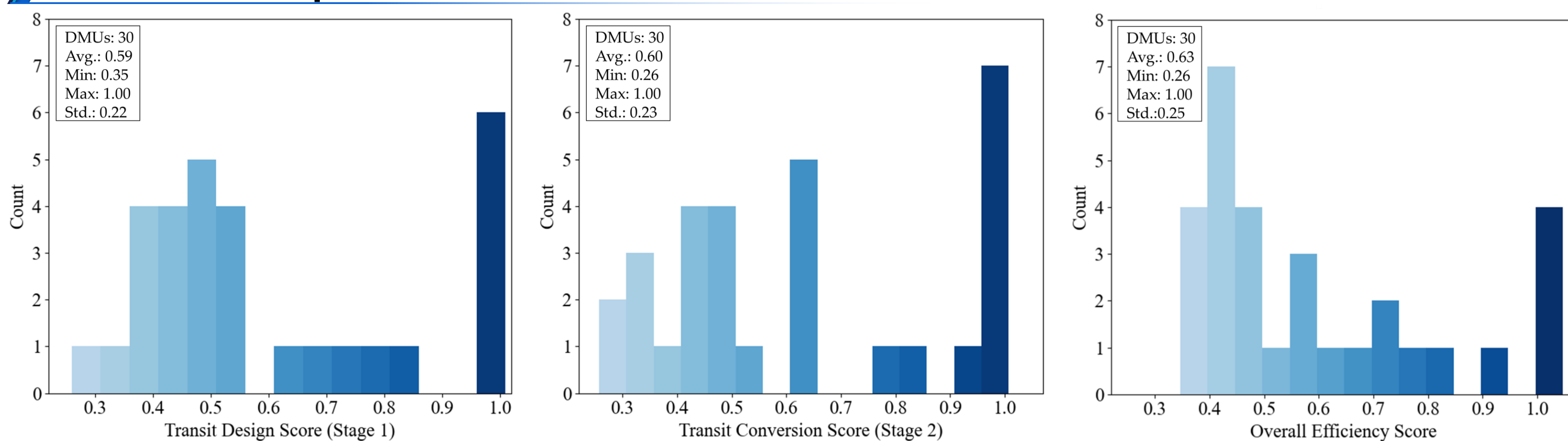
$$\theta^k = \frac{2 \times \phi^k \times \theta^k}{\phi^k + \theta^k}$$

Data Description



Result & Discussion

Results of Proposed Network DEA Model



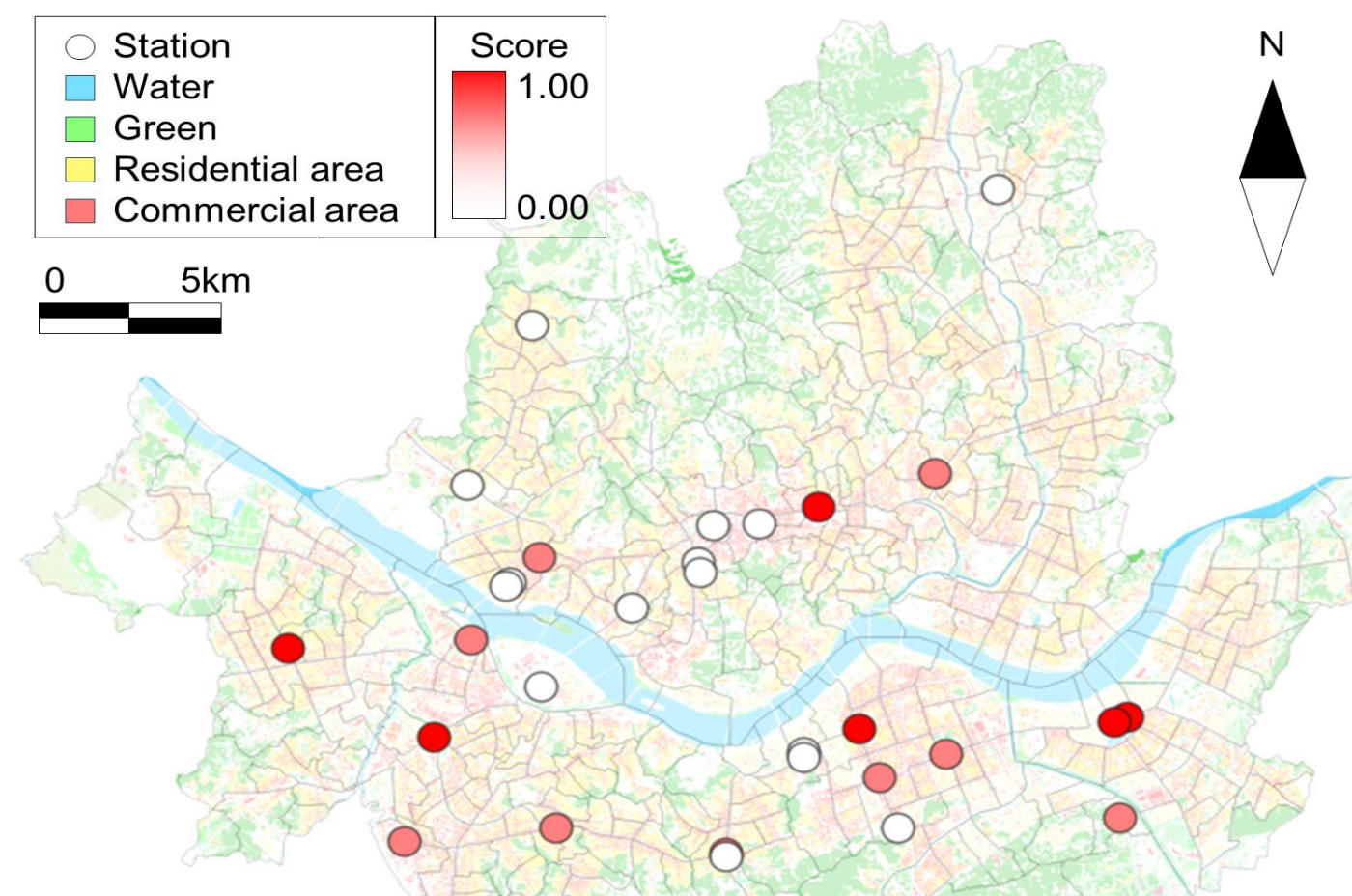
- Efficiency scores for 30 stations showed average values (Stage 1: 0.59, Stage 2: 0.60, Overall: 0.63), with several stations achieving perfect scores and many others showing substantial inefficiencies
- Among 30 stations, 4 were estimated to be efficient with an overall efficiency score of 1.00, it means that top 4 stations generated more BS trips compared to the given PT trips and BS infrastructures
- Conversely, stations that estimated to be inefficient are in urgent need of improvement

Result of Tobit Regression

	Coefficient	Std. error	z-value	p-value
Constant	-0.255	0.203	-1.254	0.210
Water area (100 m ²)	0.039	0.014	2.815	0.005
Commercial area (100 m ²)	0.013	0.005	2.834	0.005
Residential area (100 m ²)	0.009	0.004	2.154	0.031
Population (10,000 persons)	0.105	0.040	2.626	0.009
Land cost (1,000 USD/m ²)	0.018	0.009	2.036	0.042
Log-likelihood: 4.068				

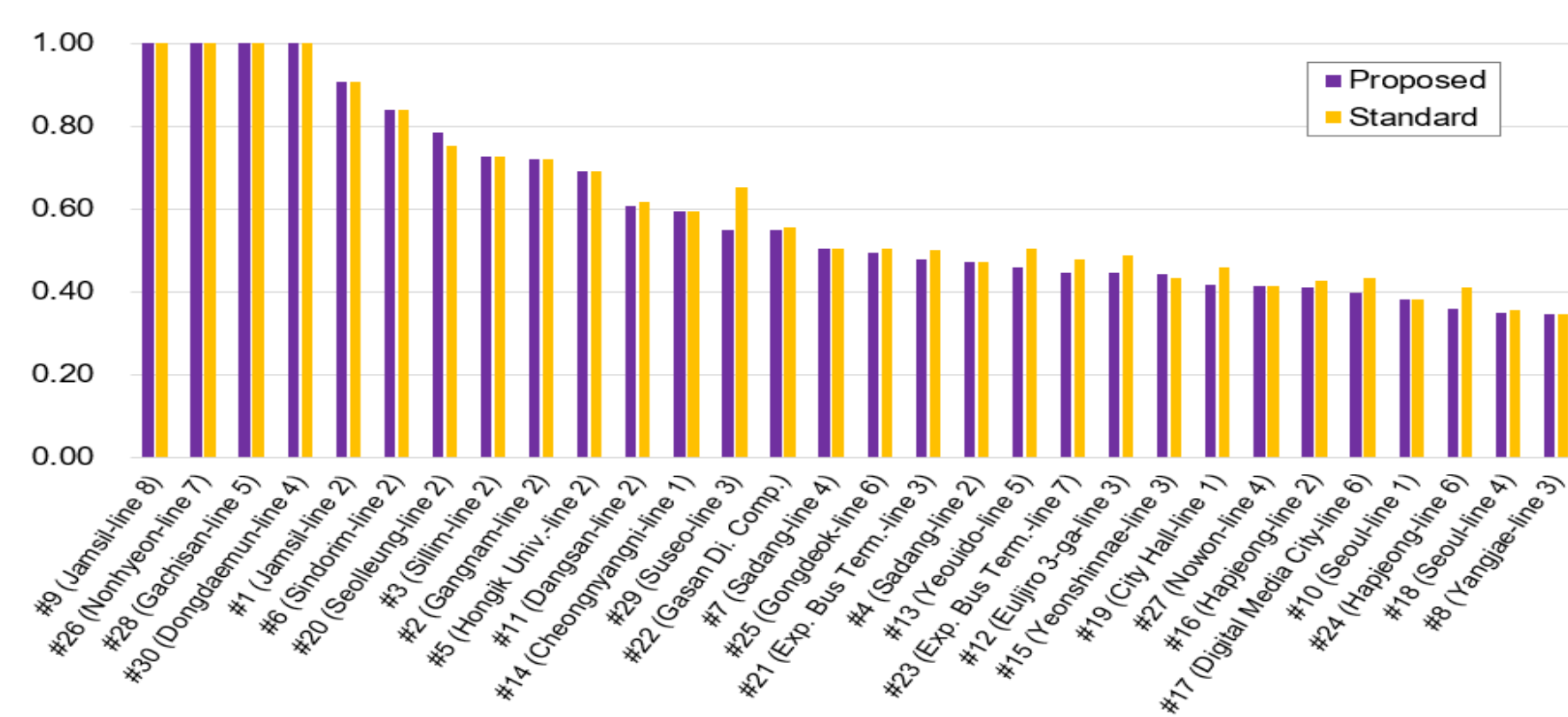
- Tobit regression identified five socio-economic factors—water area, commercial area, residential area, population, and land cost—as significant predictors of efficiency
- Higher density and mixed-use urban characteristics were associated with greater transfer efficiency

Spatial Distribution of Stations



- Stations in dense, mixed-use areas like Jamsil and Gangnam showed high efficiency, while car-oriented stations like Seoul Station and Yangjae showed low efficiency

Comparison of Efficiency Scores between Proposed and Standard Models



- The proposed model accounts for inter-stage dependency by adjusting intermediate outputs, offering a more realistic assessment
- Average efficiency score of the proposed model was slightly lower than that of the standard model (0.5925 vs. 0.6053), it provided a more stable estimation with a mean absolute percentage error (MAPE) of 3.05%

Conclusion

- This study proposed a two-stage DEA model to evaluate the intermodal transfer efficiency between public transportation (PT) and bicycle-sharing systems (BS)
- The model captures both the ability of PT infrastructure to generate transfer demand and the effectiveness of converting that demand into BS usage
- By optimizing the intermediate variable (PT trips) using first-stage efficiency scores, our model captures how inefficiencies in upstream transit infrastructure affect downstream bike-sharing usage
- This study contributes to MaaS research by providing a behaviorally grounded and methodologically robust framework for evaluating intermodal transfer efficiency
- Future research should incorporate temporal dynamics, behavioral factors, and additional modes (e.g., e-scooters) to enhance the model's adaptability and real-world applicability